

CSBIO10: Final Project

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Anxiety's Impact



**300
Million
People**

Research Question



Which genes are statistically probable to contribute to the symptoms of anxiety?

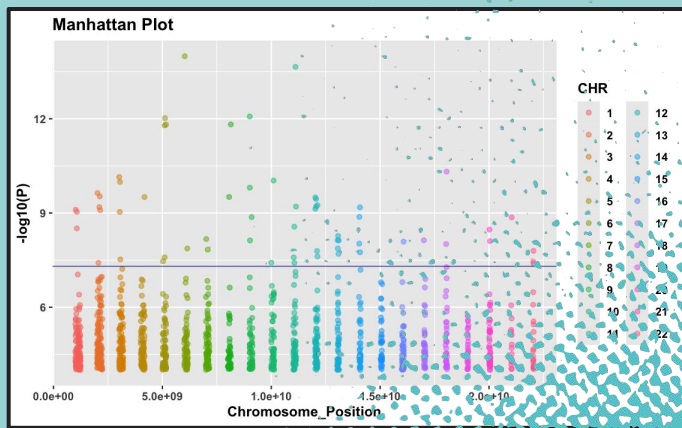
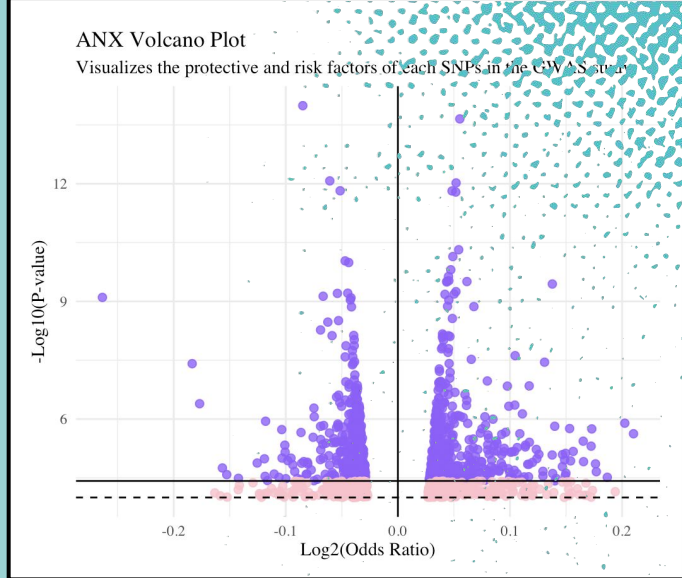
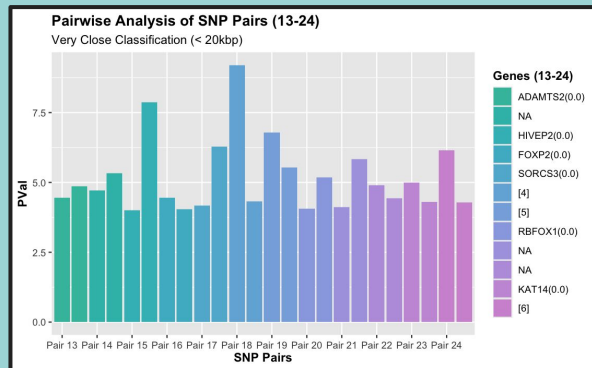
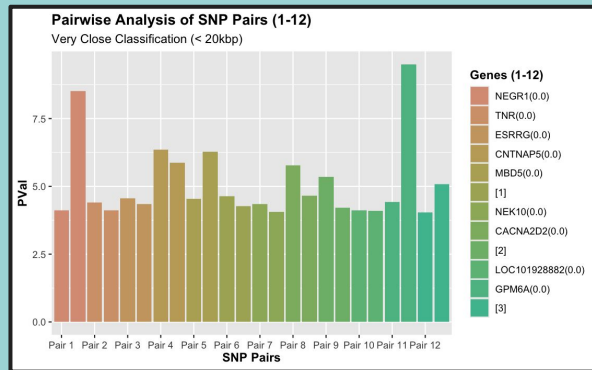
About the Dataset

- Genome-Wide Association Study (GWAS)
- Anxiety and associated SNPs



Description of Data

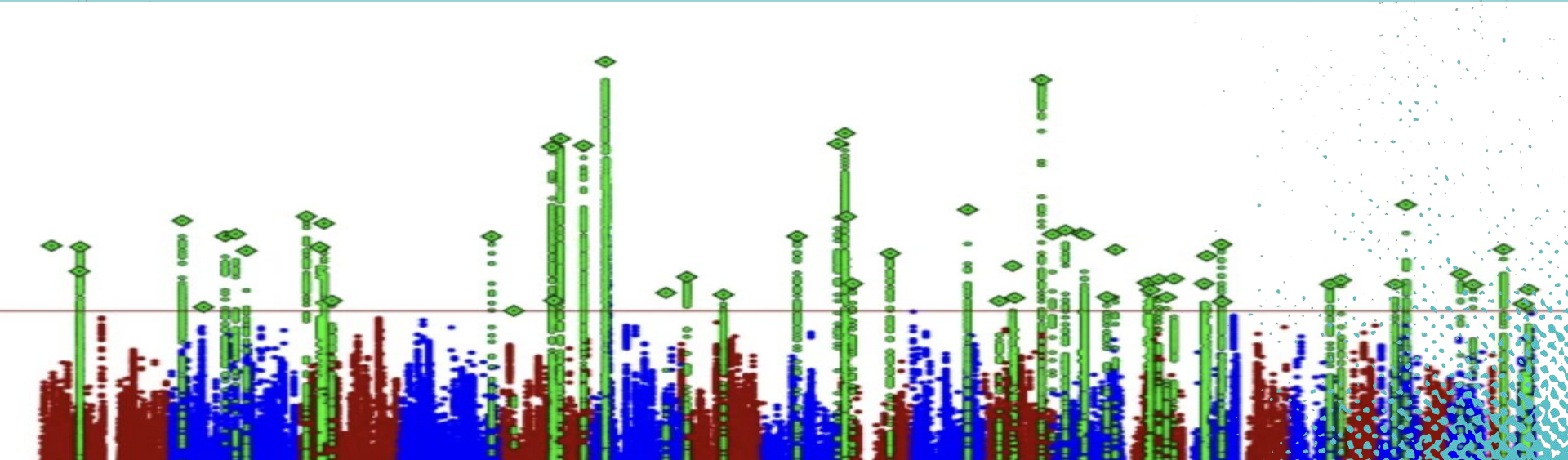
- SNPs of interest
- P-values
- Manhattan Plot
- Volcano Plot
- Bar graph



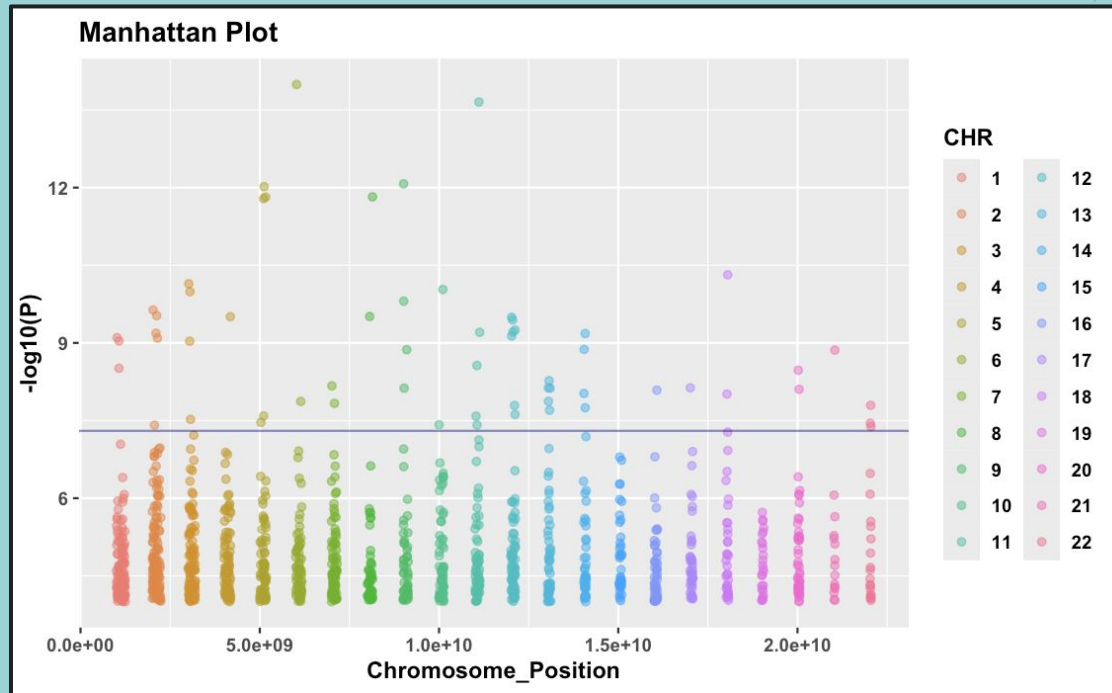
Manhattan Plot



Finding Significant SNPs



Manhattan Plot



Manhattan Plot

```
# Manhattan plot of anxiety GWAS data each dot is a SNP plotted by genome position
library('readxl')
ANX_data <- read_excel('ANX_2026_daner_fullANX_v12_woUTAH_11022026.gz.p4.clump.areator.sorted.1mhc (1).xls')
install.packages("tidyverse")
library(ggplot2)
library(tidyverse)
ANX_data %>%
  select(SNP, CHR, BP, P) %>%
  mutate(CHR = as.factor(CHR)) %>%
  arrange(CHR, BP) %>%
  #chromosome numbers have been multiplied by 100 million to separate them visually on the graph
  mutate(Chromosome_Position = as.numeric(CHR)*100000000+BP) %>%

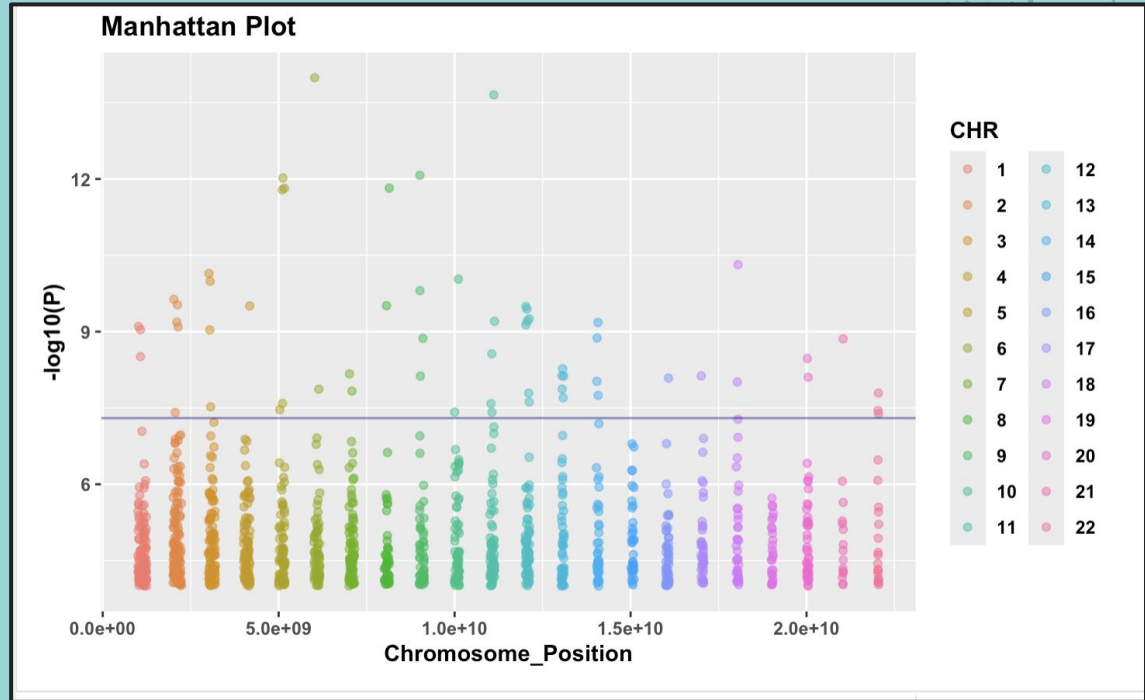
  #y intercept denotes a line of significance; from the article we pulled the data from
  #line marks the significance threshold (p < 5e-8); dots above it are significant hits.
  ggplot(aes(Chromosome_Position, -log10(P), color= CHR)) +
  geom_point(alpha=0.5) + geom_hline(yintercept = 7.30102999566, color = 'darkblue', size = 0.5, alpha = 0.5) +

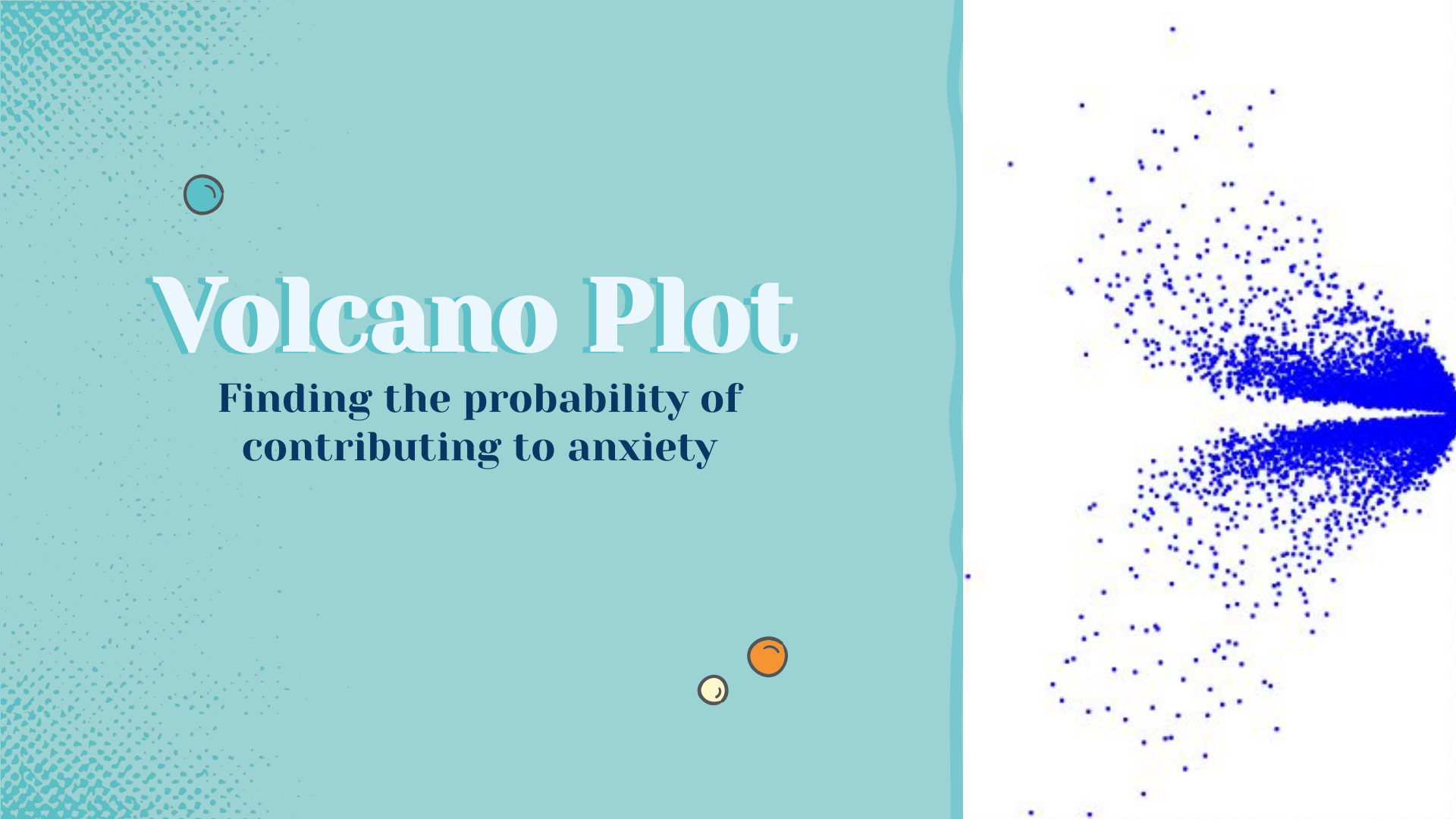
  labs(title = "Manhattan Plot") +

theme(
  plot.title = element_text(face = "bold"),
  plot.subtitle = element_text(family = "serif"),
  axis.title.x = element_text(face = "bold"),
  axis.title.y = element_text(face = "bold"),
  axis.text = element_text(face = "bold"),
  legend.title = element_text(face = "bold"),
  legend.text = element_text(face = "bold")
)
```


Main Topics

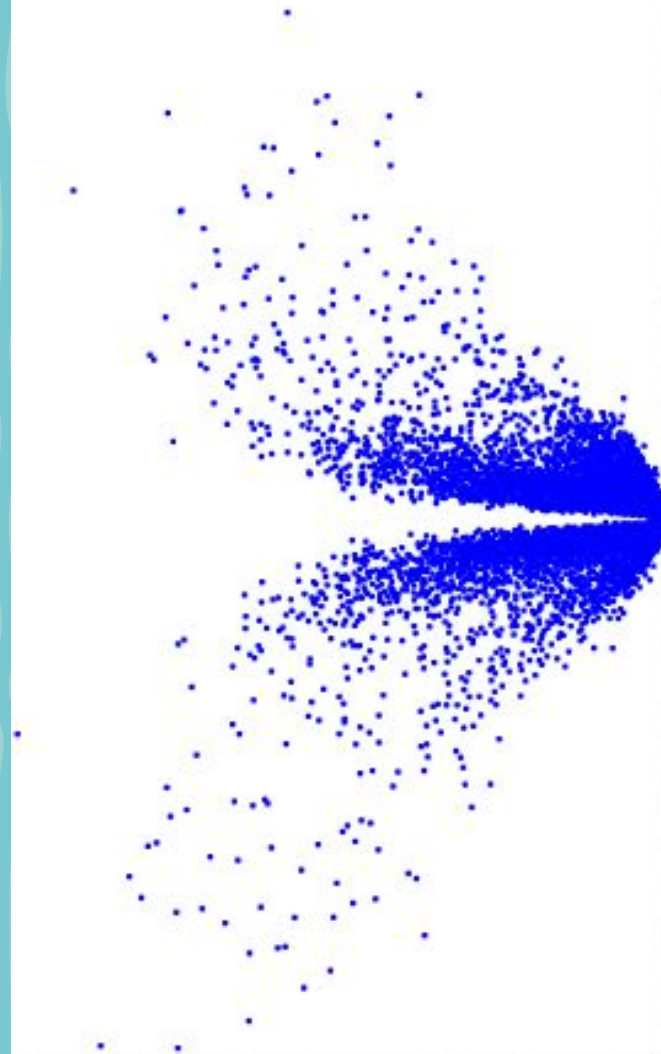
- Packages
 - ggplot2
 - Tidyverse
- Significant SNPs





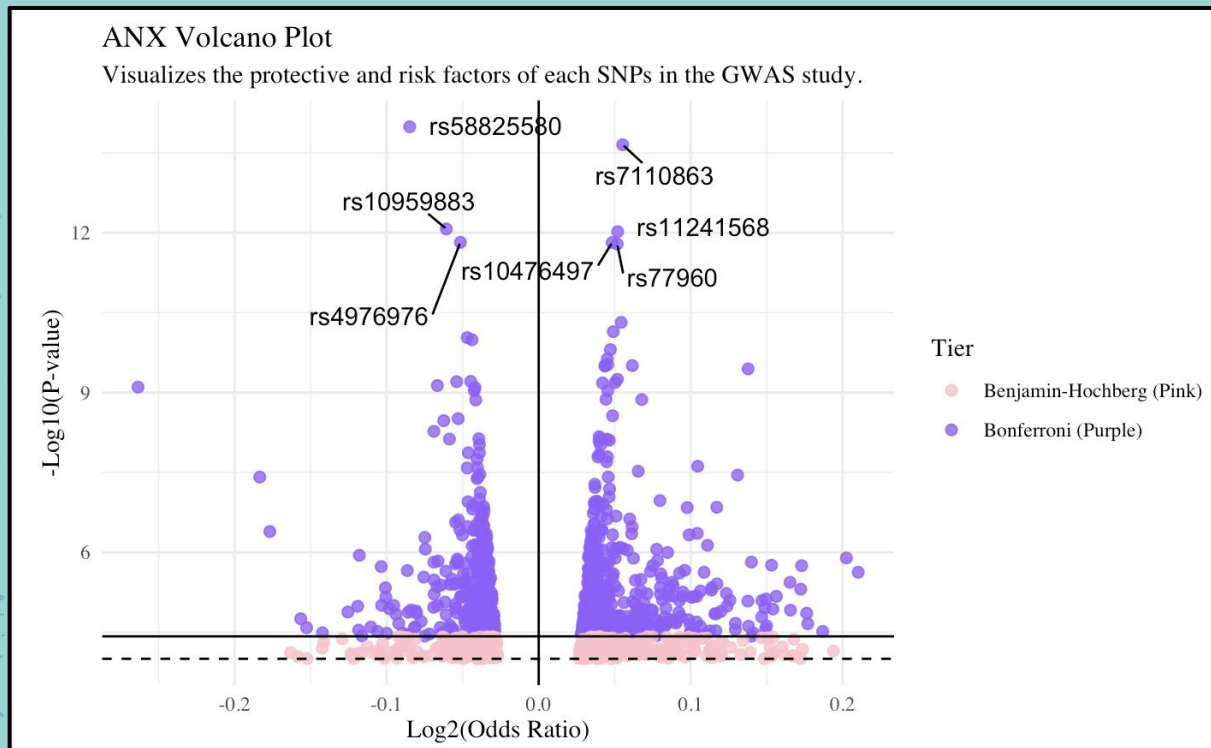
Volcano Plot

Finding the probability of
contributing to anxiety



Odds Ratio

Protective vs Risk Factors



Volcano Plot

Tiering the P-values by BH/Bonferroni

Color
Code the
BH/Bon
Pvals

```
```{r}
Install ggplot2 if you don't have it
install.packages("ggplot2")
library(ggplot2)

Step 1: Transform the the OR and PVals
Taking the log of both OR and PVals help the visualization to be more clear and discernable
ANX_dataset$logP <- -log10(ANX_dataset$P)
ANX_dataset$log2OR <- log2(ANX_dataset$OR)

Step 2: Calculate the BH/Bon thresholds (converted to -log10 scale for the Y-axis)
bh_thresh <- -log10(max(ANX_dataset$P[ANX_dataset$BH_P < 0.05]))
Look at all the raw PVals who have the BH PVals less than 0.05
The largest of those PVals will be the "worst"-most PVal that passes the BH test
bonf_thresh <- -log10(0.05 / 1320)
This is simply the Bonferonni Correction equation

Step3: Create a column to color code the 2 tiers
ANX_dataset$Tier <- "Benjamin-Hochberg (Pink)"
All rows PVals labeled as BH right now
ANX_dataset$Tier[ANX_dataset$P < (0.05 / 1320)] <- "Bonferroni (Purple)"
Now, only PVals under the Bonferroni Threshold are labeled
```

Filter the  
adjusted  
Pvals

Label the  
most  
significant  
SNPs



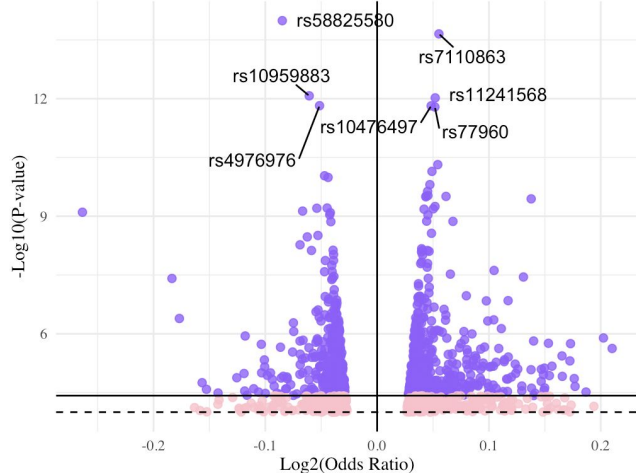
```
ggplot(ANX_dataset, aes(x = log2OR, y = logP, color = Tier)) +
 geom_point(alpha = 0.6, size = 2) +
```

```
 geom_text_repel(aes(label = Label),
 color = "black", # Labels are black
 box.padding = 0.7, # Space between labels
 max.overlaps = Inf, # All labels drawn
 show.legend = FALSE) + # Labels restricted to plot
```

```
 geom_hline(yintercept = bh_thresh, linetype = "dashed") + # Lower line
 geom_hline(yintercept = bonf_thresh, linetype = "solid") + # Upper line
 geom_vline(xintercept = 0, linetype = "solid", color = "black") + # OR divide line
 theme_minimal() +
 scale_color_manual(values = c(
 "Benjamin-Hochberg (Pink)" = "pink",
 "Bonferroni (Purple)" = "#925EFF"
)) +
 labs(title = "Volcano Plot: Significance of H/L Odds of ANX Association",
 x = "Log2(Odds Ratio)",
 y = "-Log10(P-value)",
 subtitle = "Visualizes the protective and risk factors of each SNPs in the GWAS study."
) +
```

ANX Volcano Plot

Visualizes the protective and risk factors of each SNPs in the GWAS study.



# Volcano Plot



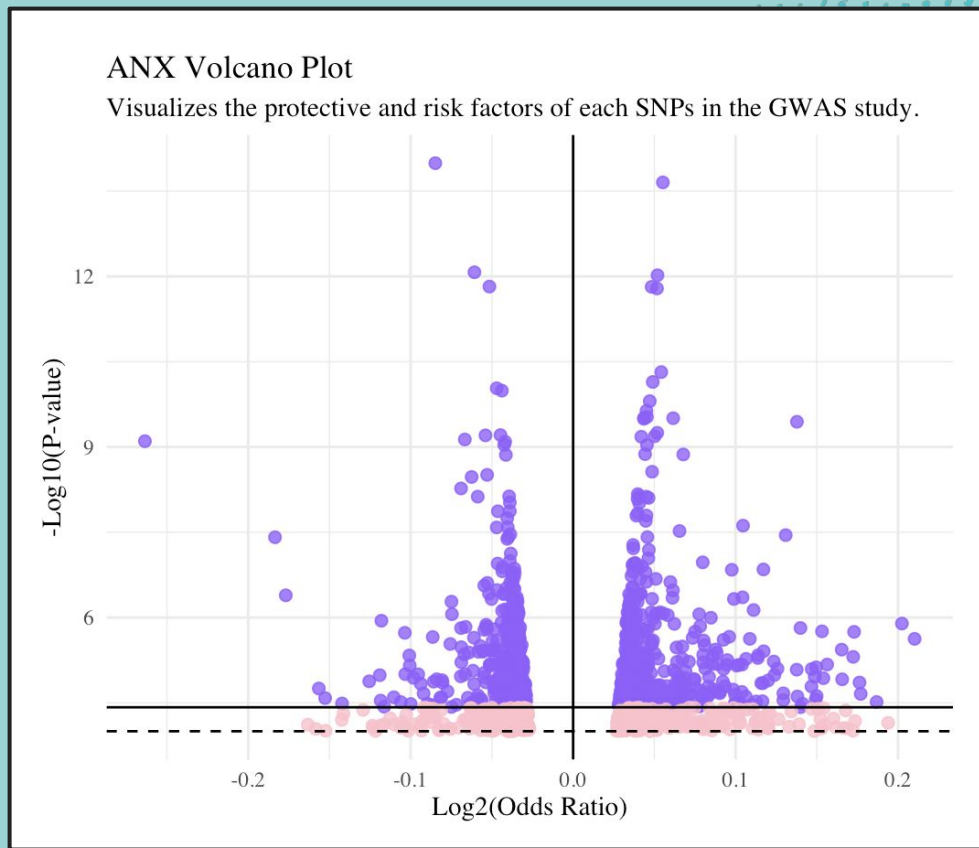
Plotting the Graph



# Main Topics



- **Packages**
  - ggplot2
  - tidyverse
  - readxl
  - ggrepel
  - Metafor
- **Benjamin-Hochberg**
- **Bonferroni**
- **Odds ratio**



# Future Work

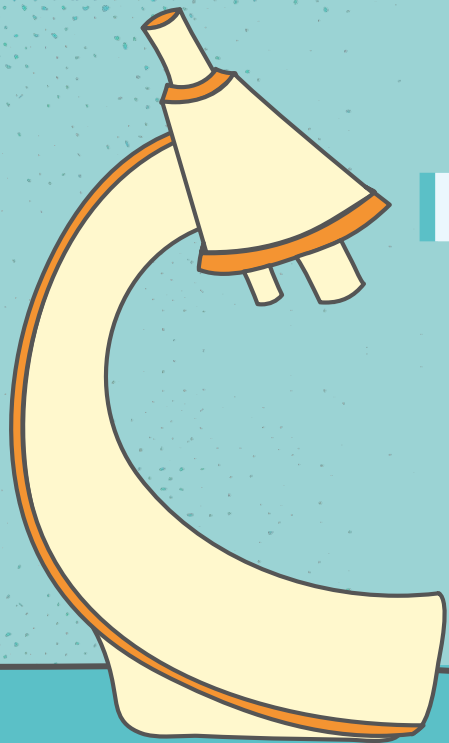


**Parent  
genome  
sequencing**

**Depression  
SNP  
Comparison**

**Cardiovascular  
SNPs**





Thanks!